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Managing boundary-spanning marketing activities for supply-chain efficiency

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Abstract The purpose of this paper is to advance and empirically investigate how perceived organisational support influences marketing managers' boundary-spanning marketing activities (cross-functional communication and customer connections), and how these activities subsequently impact supply-chain efficiency. By analysing survey responses from 348 marketing managers, we demonstrate that a supportive organisational climate provides managerial incentives not only to develop strong allegiance to the welfare of multi-foci groups within the organisation but also to engage in active cross-functional market intelligence exchange and customer servicing, which in turn leads to superior supply-chain efficiencies. The paper offers strategic guides to both academicians and practitioners on how to promote marketing managers' role in strengthening dependencies among various functional departments for cohesive product, service, and financial offerings.

Keywords market function; structure equation modelling; customer connections; market intelligence; supply-chain efficiency; cross-functional relationships

Introduction

One of the primary determinants of organisational success in a turbulent and competitive environment is supply-chain efficiency: the efficient management of the business process from product planning to distribution (Baiman, Fischer, & Rajan, 2001). Supply-chain efficiency can greatly drive down operating costs and enable shorter production period and faster distribution, thereby enhancing the productivity of the whole business. Due to slowed corporate spending in an economic downturn since 2007, supply-chain efficiency with cost containment has become one of the most critical factors in helping organisations sustain their competitive advantage.

To ensure that supply chains are finely tuned at each stage, from product fabrication to customer delivery, organisations need to align internal organisational resources with external market needs (e.g. Bettencourt, Brown, & MacKenzie, 2005; Maltz & Kohli, 1996; Moorman & Rust, 1999). The alignment of internal resources with external needs requires both (1) cross-functional and (2) customer-oriented boundary-spanning marketing activities: the former facilitates

cross-functional market intelligence exchange and the latter connects organisations with external customers.

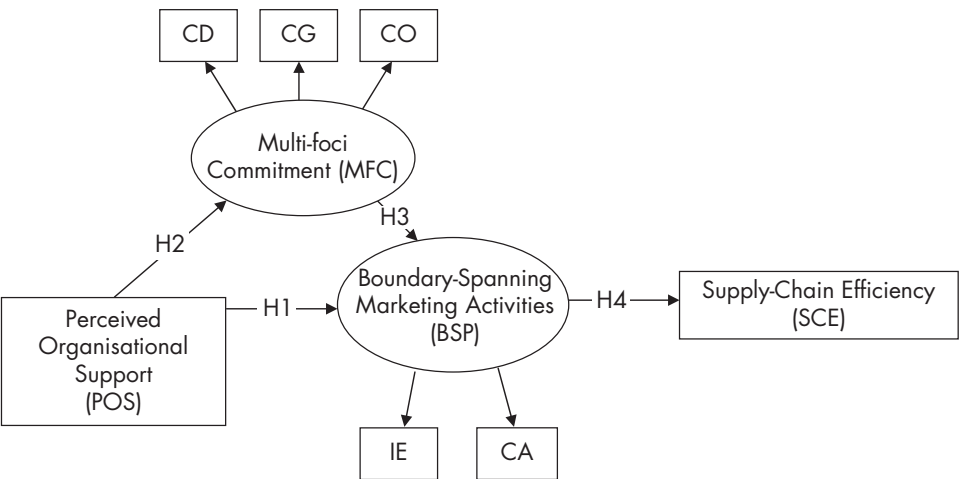
Marketing has been identified as the highest level of boundary spanning within organisations due to its cross-functional and customer-centric roles (M. McDonald, 2009). Internally, functional divisionalisation dominates the design of modern organisations based on the logic that specialised skills lead to the most efficient task fulfilment. In response, organisations are becoming highly matrixed with multiple functional areas, such as accounting, finance, marketing, and production. These functional areas are interrelated, and close collaboration facilitated by cross-functional market intelligence exchange is vital to ensure the development of common understandings of customer needs (Piercy, 2009; Sidhu & Roberts, 2008). Externally, marketing generates value through fulfilling three types of customer connections (Moorman & Rust, 1999; Verhoef & Leeflang, 2009): (1) the customer–product connection, (2) the customer–service delivery connection, and (3) the customer–financial accountability connection. These connections not only ensure that customers obtain valuable products, services, and financial information, but also generate important strategic information to guide technical development and managerial decisions regarding the daily activities of the organisation.

Although researchers commonly acknowledge the cross-functional and customer-centric roles of marketing (e.g. Day & Montgomery, 1999; Maltz & Kohli, 1996; Moorman & Rust, 1999; Verhoef & Leeflang, 2009), little is known about what factors can enhance these boundary-spanning activities and how specifically these boundary-spanning activities impact supply-chain efficiency from a managerial perspective. The objectives of this study are twofold. First, we aim to determine how perceived organisational support impacts marketing managers' internal and external boundary-spanning activities: cross-functional market intelligence exchange and customer accommodation. We suggest that perceived organisational support will have a direct effect on marketing managers' boundary-spanning activities in addition to its effect through multi-foci commitment: commitment to the marketing department, the general manager, and the organisation. Second, we aim to gain a deeper understanding from a managerial perspective of the relationship between boundary-spanning marketing activities and supply-chain efficiency. The delivery of supply-chain efficiency through technical processes, such as data capture devices, automation, and Radio Frequency Identification (RFID), has received wide attention in the literature (Lo, Yeung, & Cheng, 2009), while the empirical study on supply-chain efficiency from a managerial perspective is rather scant. Our study focuses on supply-chain efficiency from the managerial perspective and examines how marketing managers' boundary-spanning activities impact on the internal and inter-organisational supply-chain processes.

We developed a conceptual model (see Figure 1) and collected data from 348 marketing managers to test the conceptual model. The results of the structural equation model demonstrate positive relationships between perceived organisational support, marketing managers' boundary-spanning activities, and supply-chain efficiency. The results also support the partial mediating effects of multi-foci commitment on the relationship between perceived organisational support and boundary-spanning activities.

The remainder of the paper proceeds as follows. First, the 'research model' section outlines the theoretical model through the discussion of related constructs, theoretical arguments, and the hypotheses among those constructs. Second, the

Figure 1 The model of boundary-spanning marketing activities for supply-chain efficiency.



‘research method’ section explains data collection procedures and presents empirical findings about the hypothesised relationships. Finally, the ‘discussion’ section focuses on the theoretical and managerial implications of the study, as well as the limitations and possible avenues for future research.

Research model

Marketing shoulders the responsibility of both cross-functional marketing intelligence exchange and customer accommodation (e.g. Day & Montgomery, 1999; Moorman & Rust, 1999). Those boundary-spanning marketing activities play a key role in connecting internal functional areas within the organisation, gathering valuable external customer information, and communicating both internal and external information for strategic decision making. The cross-functional role of marketing requires marketing managers to cross the internal organisational boundaries due to functional separation and view the organisation as made up of closely interrelated parts (Ellinger, 2000; Maltz & Kohli, 1996). The customer-centric role of marketing requires marketing managers to cross the external organisational boundary and connect with customers (Verhoef & Leeflang, 2009). It pertains to the extent to which marketing managers are able to fulfil three kinds of customer connections (Moorman & Rust, 1999): the customer–product connection (activities connecting customers to the product design or quality issues); the customer–service delivery connection (design and delivery of ancillary actions involved in providing goods and services to customers); and the customer–financial accountability connection (activities linking marketing activities to the value of the firm).

The internal and external boundary-spanning roles of marketing – cross-functional market intelligence exchange and customer accommodation – are interrelated tasks for marketing managers, since the delivery of superior customer value relies on

successful cross-functional market intelligence exchange within the organisation. While customer accommodation helps customers access the right products, services, and financial information, the connectivity in these aspects can be achieved only through efficient market intelligence exchange among all interrelated organisational functional departments. In our paper, we suggest that perceived organisational support facilitates marketing managers' active boundary-spanning roles, which will in turn enhance supply-chain efficiency.

Perceived organisational support

Perceived organisational support reflects employees' beliefs regarding how much their respective organisations value their contributions and care about their well-being (Eisenberger, Huntington, Hutchiso, & Sowa, 1986). The construct implies that organisations that reward employees' increased work efforts and promote employees' welfare demonstrate humanlike characteristics: they are perceived as being caring, supportive, and nurturing in the eyes of their employees (Eisenberger, Fasolo, & Davis-LaMastro, 1990; Rhoades & Eisenberger, 2002; Settoon, Bennett, & Liden, 1996). According to social exchange theory, employees will reciprocate by responding to the caring and supportive organisations with positive attitudes and better job performance (Blau, 1964). Research on perceived organisational support widely acknowledges the reciprocity between employees and organisations (e.g. Armstrong-Stassen & Ursel, 2009; Rhoades & Eisenberger, 2002). Research findings demonstrate that perceived organisational support leads to various desirable job-related outcomes, such as employees' job satisfaction, positive mood, reduced stress, increased commitment towards the organisation, and reduced employee turnover (Armstrong-Stassen & Ursel, 2009; Rhoades & Eisenberger, 2002).

However, no prior research has examined the relationship between marketing managers' boundary-spanning activities and their perceived organisational support. Research on the impact of perceived organisational support on boundary-spanning behaviours has focused most commonly on salespeople, service, and repair technicians. For example, Bahakus, Cravens, Johnston, and Moncrief (1996) found that perceived organisational support positively impacts sales people's intrinsic motivation, job satisfaction, and fulfilled expectations. Piercy, Cravens, Lane, and Vorhies (2006) found that perceived organisational support directly impacts sales people's organisational citizenship behaviour and indirectly impacts sales people's in-role behavioural performance.

Marketing managers' perception of organisational support reflects their organisations' readiness to provide them with assistance when they are fulfilling marketing-related job responsibilities and to reward their increased efforts in carrying out cross-functional and customer-centric roles. A high level of perceived organisational support indicates that organisations are always ready to provide aids to facilitating job performance and dealing with stressful situations. When marketing managers are assured that aids from the organisations are always available to support their boundary-spanning activities, they will be highly motivated to devote themselves to exchanging market intelligence with organisational members from other functional departments and accommodating customers' needs.

Such a consequence of perceived organisational support is consistent with prior research and social exchange theory, which suggests that a social exchange

relationship is developed between two parties with reciprocal obligation in each party (Aselage & Eisenberger, 2003; Settoon et al., 1996). Based on the reciprocity norm, when the organisation rewards their increased work effort and cares about their socio-emotional needs, marketing managers develop an emotional obligation to the organisation's welfare and goals. The caring and respect implied by perceived organisational support provides marketing managers with psychological assurance so that they will feel encouraged to engage in boundary-spanning activities. Therefore, we suggest that marketing managers who perceive organisations to be supportive are more likely to reciprocate with increased efforts to exchange cross-functional market intelligence and accommodate customers' needs:

H1: Perceived organisational support of marketing managers positively impacts their boundary-spanning activities of cross-functional market intelligence exchange and customer accommodation.

Multi-foci commitment

Organisational commitment has been conceptualised as the relative strength of an individual's identification with and involvement in a particular organisation (Steers, 1977). The definition views commitment to be a global construct (targeting the organisation as a whole) rather than a constituency-specific construct. A number of authors, however, have found that employees' commitment to the workplace is multi-foci (Becker, 1992; Stinglhamber, Bentein, & Vandenberghe, 2002). The foci of commitment refer to the individuals and groups to whom an employee is attached. A number of foci, including co-workers, superiors, subordinate, and other groups or individuals collectively comprising the organisation, are salient (Becker, 1992). Since marketing managers interact with different groups at multiple levels, from department settings to the organisation as a whole, it is not surprising that they may demonstrate distinct attachment to different groups whose opinions matter to them. The process of marketing managers' internalisation of the structural roles they play in the organisation leads to their multi-foci commitment.

Multi-foci commitment of marketing managers ranges from their commitment to immediate/local foci, such as the marketing department to global/organisational foci, reflected in their attachment to the top management and to the organisation. Specifically, the paper examines three foci of marketing managers' commitment: commitment to the marketing department, commitment to the general manager, and commitment to the organisation.

Multi-foci commitment captures marketing managers' loyalty towards the marketing department, the general manager, and the organisation. A high level of perceived organisational support has long been related to increased organisational commitment (Rhoades, Eisenberger, & Armeli, 2001; Shore & Wayne, 1993). Researchers suggest that employees interpret the treatment they receive from the organisation as the organisation's commitment to them (Wayne, Shore, & Liden, 1997). Based on the reciprocity rule of social exchange theory, employees' perception of the organisation's commitment to their welfare will have significant impact on their own commitment to the organisation (Settoon et al., 1996).

Drawing upon compelling empirical support (e.g. Aselage & Eisenberger, 2003; Rhoades & Eisenberger, 2002), we argue that the positive affective state associated

with perceived organisational support will foster marketing managers' multi-foci commitment. Committed marketing managers demonstrate a sense of belonging and identification that increases their involvement in not only their functional tasks but also the organisational welfare. In contrast, when marketing managers do not feel respect and caring from the organisation, or when they think that their increased work efforts do not receive proper rewards, they will experience psychological tension or distress and develop a negative affective state towards their functional tasks, the general manager, and the organisation. Thus we expect:

H2: Perceived organisational support positively impacts marketing managers' multi-foci commitment, which reflects their commitment to the marketing department, to the general manager, and to the organisation.

Multi-foci commitment reflects marketing managers' favourable attitudes and their strong sense of attachment to the marketing department, the general manager, and the organisation. Research demonstrates that not only committed employees are more likely to fulfil their in-role tasks in a dependable fashion but they also demonstrate pro-social and extra-role behaviours (Bettencourt et al., 2005). For example, Bettencourt et al. (2005) found that organisational commitment of frontline service employees directly impacts three dimensions of their customer-oriented boundary-spanning behaviours: external representation (advocating the organisation's products and services to customers); internal influence (communicating with co-workers to improve service offerings); and service delivery (serving customers in a flexible and dependable manner). Therefore, we suggest that multi-foci commitment creates an internalised psychological obligation for marketing managers to work proactively for the best interests of the organisation through their boundary-spanning activities – disseminating important market intelligence across functional departments, translating customer needs into product specifications, linking customer needs to the operations of frontline employees, and providing financial justification for marketing activities:

H3: Multi-foci commitment positively impacts marketing managers' boundary-spanning activities of cross-functional market intelligence exchange and customer accommodation.

Supply-chain efficiency

The supply chain involves a set of internal and inter-organisational processes, which extend from planning to product distribution (Piercy, 2007). Supply-chain efficiency is generated not only through internal organisational efficiency, but also through efficiency on handling both the supply and the demand sides of relationships. The efficiency improvements in supply chains enable organisations to generate clear financial benefits through cost reduction, while maintaining the quality of product and service offerings to customers (Sethi & King, 1994).

The coordination and control of both internal and inter-organisational processes in supply chains, including inbound logistics, manufacturing, outbound logistics, and service, is inherently cross-functional and customer-centric, which requires boundary-spanning marketing efforts (Philip, 2007; Piercy, 2007). Marketing managers' efficient boundary-spanning activities help to break down functional and divisional

boundaries and facilitate cross-functional connectedness and collaboration, which is prerequisite before cross-functional trust and participation can come into play. Since organisational members with different functional backgrounds are equipped with unique experiences, it is essential for marketing managers to communicate important customer information effectively in order for organisational members to develop a common understanding of and a shared goal regarding customer needs. The cross-functional synergy and the common understanding of customer needs enable the organisation to provide better products, reliable service, and optimal financial solutions to customers, which are the basis of long-term healthy supply-chain relationships (Hammervoll, 2009; Leemon 1995).

Boundary-spanning marketing activities that break cross-functional barriers within the organisation and connect the organisation to the customers also promote information flow through the supply chain (Battor & Battor, 2010). Efficient information flow is critical for business areas such as production forecasts and inventory management. For example, Ellinger (2000) found that boundary-spanning marketing activities enable optimised distribution and delivery. Without those activities, organisations may produce poorly conceived products or services and fail to generate appropriate volume to meet current or future customer needs. Specifically, marketing managers' poor performance of cross-functional market intelligence exchange and customer accommodation may leave the organisation with little knowledge about what kinds of product need to be made and when products or services need to be delivered. Failure to balance business capacity levels and sales requests or forecasts properly will also cause waste in production and increased cost in inventory management, thus hindering supply-chain efficiency. Therefore, we hypothesise:

H4: Marketing managers' boundary-spanning activities have a positive relationship with supply-chain efficiency.

Research method

Pilot study

A survey provided the primary means for data collection. We conducted a pilot study before the data collection to detect problems that may pose threats to the construct validity of the scales used in the questionnaire. Specifically, we conducted preliminary in-depth interviews with five marketing managers and 10 EMBA students with work experience as marketing managers. The in-depth interviews ensured that the survey instrument was appropriate for marketing managers and covered all relevant issues related to the research question—marketing managers' roles in enhancing supply-chain efficiency. The in-depth interviews were conducted in person and were framed around the basic research questions, including marketing managers' general comments about their perceptions of organisational support, their boundary-spanning roles in the organisations, and the efficiency of internal and inter-organisational processes. Each interview lasted around one-and-a-half hours.

Interviewees provided useful feedback on their comprehension and the relevance of the survey questions. They also commented on the appropriateness of the

survey's use of certain terminology and identified any other problems they might have found in the study. Their input provided a basis for the development of our survey questionnaire and ensured the construct validity of the scales used in the questionnaire.

Sample and data collection procedure

The sample of the study was drawn from the membership roster of a professional managerial association in the United States. We randomly selected a master list of around 3000 marketing managers and collected their e-mail addresses from the membership roster. We then set up an online survey through a commercial survey support site to collect data. We gathered the data in two successive e-mail invitations. A personalised e-mail with instructions and a link to the online survey was sent to each manager in the first round of e-mails. As an incentive for participating, the respondents were informed that, in return for completing the survey, they would receive a general report of the results. We sent out a second round of e-mails to those respondents who had not replied two weeks after the first round of e-mails. A total of 355 respondents completed and submitted the survey after two rounds of e-mails, yielding a response rate of 11.8%. Valid responses came from 348 marketing managers, since seven responses were missing too much information and were unusable. Approximately 41% of the 348 surveys were completed by males and the rest by females. The average length of time respondents had been in their organisations was approximately 6.4 years, with a range from 1 year to 31 years. More detailed profiles of the respondents are given in Table 1.

To assess the possibility of non-response bias, we used the extrapolation procedure by Armstrong and Overton (1977). Specifically, we split the data into two parts according to the date of online survey submission. The first 75% responses (261 responses) were classified as 'early' and the last 25% (87 responses) as 'late' respondents. We used the latter 25% to represent non-respondents. *T*-tests were applied to respondents' background information – experience (years in the position), age, and companies' gross revenue – to reveal any significant differences between the early and the later respondents. No significant differences were found between the two groups on any of these variables, indicating that non-response bias was not a problem for the analysis.

Measures

We used seven-point Likert scales to measure the constructs in the study. The seven-point scales were anchored at each end with the terms 'strongly disagree' or 'strongly agree'.

Following Rhoades et al. (2001), we measured perceived organisational support with eight items. The scale captured the organisation's readiness to reward increased work effort and to care about marketing managers' well-being. The following is a sample item: 'The organisation really cares about my well-being'.

While established scales are available for commitment at the organisational level, there is no existing scale to measure commitment to the marketing department and commitment to the general manager in particular. We adapted Allen and Meyer (1990)'s scale of organisational commitment so that the items addressed commitment related to three foci: the marketing department (six items), the general

Table 1 Profile of final sample of 348 managers (130 male, 41.0%; 187 female, 59%).

	Count	%
<i>Company gross revenue (US \$)</i>		
<100 million	125	41.4%
100 million–500 million	53	17.5
500 million–1 billion	21	7.0%
1 billion–1.5 billion	14	4.6%
1.5 billion–20 billion	21	7.0%
>20 billion	68	22.5%
Total	302	
<i>Years in the company</i>		
≤2	75	23.7%
2–4	59	18.7%
4–6	65	20.6%
6–10	49	15.5%
≥10	68	21.5%
Mean	6.4	
Median	4	
Minimum	1	
Maximum	31	
<i>Years as a marketing manager</i>		
≤1	55	17.4%
1–2	86	27.2%
2–3	73	23.1%
3–5	63	19.9%
>5	39	12.3%
Mean	3.5	
Median	3	
Minimum	1	
Maximum	23	
<i>Age</i>		
≤29	46	14.4%
30–34	66	20.7%
35–39	53	16.6%
40–44	56	17.6%
45–49	43	13.5%
≥50	55	17.2%
Total	319	

manager (six items), and the organisation (six items). The following are sample items for commitment to the marketing department: ‘I really feel as if the marketing department’s problems are my own’; commitment to the general manager: ‘I am

willing to put a great deal of effort beyond that normally expected to help the general manager'; and commitment to the organisation: 'I would like to continue working for this organisation'.

Consistent with our conceptualisation, we measured both internal and external boundary-spanning activities of marketing managers. The internal boundary-spanning activities capture marketing managers' efforts of sharing market intelligence and interacting with organisational members from different functional departments. We adapted Jaworski and Kohli (1993)'s scale on interdepartmental connectedness to measure cross-functional market intelligence exchange. The following is a sample item: 'I find communication between the marketing department and other departments easy'. The external boundary-spanning activities capture marketing managers' effort of accommodating customers' needs. Specifically, the scale focuses on how well marketing managers provide their customers with products, services, and financial solutions. We derived a three-item scale to measure customer accommodation from Moorman and Rust (1999)'s study. The following is a sample item: 'The firm's ability to translate customer needs into technical specifications for new products/services resides in me'.

Supply-chain efficiency reflects the organisation's capability of controlling and coordinating both internal and inter-organisational business processes. To avoid common method bias, we measured supply-chain efficiency in two ways (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003). First, we developed a 10-item scale to measure supply-chain efficiency. The scale focuses on the organisation's capability of handling both the internal and inter-organisational business processes from planning to product distribution. The following is a sample item: 'Our firm has the capability to manage its overall supply chain efficiently'. Second, we collected the organisation's return on assets (ROA) information through COMPUSTAT to capture supply-chain efficiency. ROA reflects how effective a company's assets are in generating revenue, which is a commonly used logistics financial performance measure (Roth, Cattani, & Froehle, 2008). We developed the company's supply-chain efficiency index through combining both the normalised survey responses and ROA data.

We assessed all scales in terms of item loadings, reliability, convergent validity, and discriminant validity before hypothesis testing. Nearly all factor loadings exceeded the recommended threshold of .70 (Hair, Anderson, Tatham, & Black, 1998). Two items showed loadings between .65 and .70. These values are acceptable, since other items related to the same constructs have higher reliability scores (Hair et al., 1998). All loadings are significant at $p < .01$. Therefore, the overall reliability of the items is acceptable.

We checked both Cronbach's alpha and the composite reliability as measures for convergent reliability. Fornell and Larcker (1981) have argued that the composite reliability measure is superior to Cronbach's alpha, since it is a measure of the overall reliability of a collection of items in the model. All composite reliability estimates in the study are above the .70 threshold suggested by Hair et al. (1998). We also checked the AVE of all scales. Fornell and Larcker (1981) suggest that an AVE above .50 shows convergent validity. All AVE estimates in the study are above the threshold. Both composite reliability and AVE estimates support the convergent validity of the measures.

We checked discriminant validity of the scales through two approaches. First, we followed the guidelines set by Chin (1998). We checked the item loadings of the scales and found that the scales all load more strongly on their own construct

than on the other construct. We then compared pair-wise chi-square statistics of any two scales: one where the correlation between the two constructs is set free and the other where the correlation between the two constructs is constrained to 1. All chi-square statistics in the unconstrained model are significantly lower than the chi-square statistics in the constrained model ($p < .01$). We also checked the discriminant validity of the scales with AVE method as proposed by Fornell and Larcker (1981). To assure discriminant validity, Fornell and Larcker (1981) suggest that the square root of AVE for a construct should be greater than the absolute value of the standardised correlation of the construct with any other construct in the study. Table 2 shows the square root of AVE, means, standard deviations, and intercorrelations between the scales. As presented in Table 2, the diagonal elements of square root of AVE for constructs are all larger than any other standardised correlations. Therefore, both approaches supported the discriminant validity of the scales.

Structural equation analysis

We conducted a structural equation model with AMOS 16.0 in SPSS and used composite indicators for multi-foci commitment and boundary-spanning marketing activities to simplify the structural model (Bollen, 1989). Before conducting the structural equation model analysis, we checked the normality of the data by examining kurtosis statistics for the scales and their standard error. Kurtosis statistics for all scales were found to be less than twice their standard error, which demonstrates that normality was not a problem for the analysis (Tabachnick & Fidell, 1996). We also checked Mahalanobis distance and found that there is no outlier in the data set (Johnson & Wichern, 2007). Furthermore, we conducted a

Table 2 Descriptive statistics and correlations among the constructs^a.

Construct	Mean	SD	POS	CD	CG	CO	IE	CA	SCE
Perceived Organisational Support (POS)	4.92	1.27	[.82]						
Multi-foci Commitment (MFC)									
Commitment to the Marketing Department (CD)	5.57	1.17	.53 ^b	[.76]					
Commitment to the General Manager (CG)	4.38	1.43	.56 ^b	.46 ^b	[.83]				
Commitment to the Organisation (CO)	5.13	1.36	.72 ^b	.62 ^b	.60 ^b	[.80]			
Boundary Spanning Performance (BSP)									
Cross-functional Market Intelligence Exchange (IE)	5.89	.82	.36 ^b	.33 ^b	.26 ^b	.30 ^b	[.73]		
Customer Accommodation (CA)	4.08	1.36	.15 ^b	.14 ^b	.04	.13 ^c	.15 ^b	[.81]	
Supply Chain Efficiency (SCE)	4.65	1.21	.54 ^b	.34 ^b	.40 ^b	.53 ^b	.34 ^b	.18 ^b	[.78]

^aThe square root of the average variance extracted (AVE) are shown in parentheses on the diagonal. A total of 348 marketing managers participated in the study. The items corresponding to each construct/dimension were summed and averaged in order to obtain a summated index. The summary statistics are reported for this index. ^bCorrelations have $p < .01$; ^ccorrelations have $p < .05$.

test of missing completely at random (MCAR) to evaluate the randomness of missing data (Hair et al., 1998). The test showed an insignificant result ($p > .05$), which demonstrated that missing values were randomly scattered. Since the proportion of the responses with missing values was small, we followed Little and Rubin (1987)'s recommendation of using a maximum likelihood estimation method for analysis of the data with missing values.

Our study follows two general suggested guidelines for assessing structural models, indicated by CFI (comparative fit index) and RMSEA (root mean squared error of approximation). For a 'good' model, CFI should be $>.90$ and desirable RMSEA $<.05$ (R.P. McDonald, 1999). Our results indicated that the data fit the structural equation model satisfactorily ($\chi^2 = 18.75$, $df = 12$, $NFI = .98$, $CFI = .99$, $GFI = .99$, $RMSEA = .04$) (see Table 3).

More specifically, our results showed that perceived organisational support significantly impacts boundary-spanning marketing activities ($\beta = .32$, $p < .01$). The findings supported H_1 and demonstrated that marketing managers are willing to exert more efforts in cross-functional market intelligence exchange and customer accommodation in organisations that are more supportive of their boundary-spanning roles. We also found highly significant relationships between perceived organisational support and marketing managers' multi-foci commitment ($\beta = .81$, $p < .00$). It showed that marketing managers who perceive a higher level of organisational support are more committed to their functional department, the

Table 3 The results of the structural models.

	Standardised parameter estimates (<i>t</i> -values)
Structural model	
POS $\rightarrow$ BSP	.32(2.71)
POS $\rightarrow$ MFC	.81(19.59)
MFC $\rightarrow$ BSP	.52(3.92)
BSP $\rightarrow$ SCE	.74(7.04)
Measurement model	
MFC $\rightarrow$ CD	.68(13.91)
MFC $\rightarrow$ CG	.68(13.64)
MFC $\rightarrow$ CO	.90*
BSP $\rightarrow$ IE	.48*
BSP $\rightarrow$ CA	.23
Goodness of fit statistics	
Chi-square	18.75
df	12
NFI	.98
CFI	.99
GFI	.99
RMSEA	.04
R^2 for SCE	.55

Notes: All parameter estimates are significant at $p < .01$. *The unstandardised coefficient corresponding to this parameter was set to equal 1.00 to fix the scale of the latent variable.

general manager, and the organisation. These findings supported H₂. There is also a highly significant positive relationship between multi-foci commitment and boundary-spanning marketing activities ($\beta = .52, p < .00$). The findings supported H₃ and indicated that marketing managers who committed to the marketing department, the general manager, and the organisation engage in more activities of cross-functional market intelligence exchange and customer accommodation. As predicted, there is a significant positive relationship between marketing managers' boundary-spanning activities and supply-chain efficiency ($\beta = .74, p < .00$). Therefore, H₄ was supported.

To summarise, the findings demonstrated the robustness of the structural model and all four hypotheses were validated. The results demonstrated that perceived organisational support and multi-foci commitments promote marketing managers' boundary-spanning marketing activities, which in turn strengthen the control and coordination of internal and inter-organisational supply-chain processes.

Discussion

Academic implications

Two research questions drove this study: (1) What are the key factors that promote marketing managers' both internal and external boundary-spanning activities? (2) How do marketing managers' boundary-spanning activities impact supply-chain efficiency? The study addressed these questions through the development and empirical testing of a theoretical model that places both marketing managers' internal (cross-functional market intelligence exchange) and external (customer accommodation) boundary-spanning activities as central points. The empirical results from 348 marketing managers supported our research model in general and provided strong support for each of the hypotheses. Our findings demonstrated that the relationship between perceived organisational support and marketing managers' boundary-spanning activities will be enhanced by a higher level of marketing managers' commitment to the marketing department, the general manager, and the organisation. Furthermore, supply-chain efficiency is delivered in organisations where marketing managers are active in cross-functional market information sharing and customer accommodation.

Managerial implications

Our study offers two important managerial implications. First, a supportive organisational environment that provides respect and psychological assurance is necessary before marketing managers take the initiative to shoulder the responsibility of breaking any internal functional barriers and accommodating customers' needs. Marketing managers have as their responsibility the sharing of information with both customers and managers from other functional departments. Perceived organisational support explains why some marketing managers are committed to their boundary-spanning tasks but others are not. Favourable treatment and socio-emotional assurance from the organisation are important to help marketing managers develop an active interest in and a strong commitment to the welfare of the marketing department, the general manager, and the organisation. This commitment is

prerequisite to the successful execution of challenging boundary-spanning tasks. Furthermore, whereas previous research has examined the influence of perceived organisational support on organisational commitment (commitment targeting at the organisation as a whole), our study showed that marketing managers' multi-foci commitment (commitment to the marketing department, the general manager, and the organisation) is a more accurate predictor of the behavioural capabilities of marketing managers, since multi-foci commitment better captures marketing managers' attachment to different working groups. Only when organisations value marketing managers' efforts in boundary-spanning tasks will marketing managers show strong allegiance to their functional department, the general manager, and the organisation. Those highly committed marketing managers will exert more effort to disseminate market intelligence within the organisation and to leverage customer information to create value.

Second, there is a strong link between marketing managers' boundary-spanning activities and supply-chain efficiency. Marketing managers who are capable of exchanging market intelligence across different functional departments within the organisation and connecting with customers can ensure organisation-wide responsiveness to the 'market pull' and can create strong cross-functional and inter-organisational synergy, which are prerequisites of supply-chain efficiency. The important findings of how marketing managers' boundary-spanning activities impact supply-chain efficiency also shed new light on how to leverage managerial resources in generating unique long-term organisational competitive advantages in the marketplace. As Kiggundu and Jorgensen (1983) have suggested in their review of comparative management research in administrative theory, technical methods can be easily replicated, but managerial activities that span the intra-organisation and the organisation–environment interface are much more difficult to transfer.

Conclusion

Our study provides valuable information on how to manage boundary-spanning marketing activities for supply-chain efficiency. Achieving supply-chain efficiency involves more than just technical monitoring of the physical flow of products. More importantly, special managerial attention must be paid to the active role of personnel involved in the process, especially marketing managers who are shouldering the responsibility of coordinating internal and external information flow in supply chains. Supply chains span multiple business areas in the organisation such as order management, production, and distribution. Critical decisions involved in those business areas are dependent on accurate information from both the functional departments and the customer marketplace. Marketing managers' boundary-spanning activities that contribute to information gathering and sharing within and beyond the organisation can enhance the efficiency in supply-chain processes.

Enhancing marketing managers' boundary-spanning roles is an imminent task for all organisations, since managerial resources are critical for long-term organisational competitive advantages in the marketplace (Brownlie, Hewer, & Ferguson, 2007). Top management teams in organisations need to assign resources and devote managerial support to facilitate marketing managers' cross-functional and customer-oriented activities, since boundary-spanning efforts are always challenging (Vafeas, 2010). The boundary-spanning efforts of marketing managers help to spread

marketing philosophy across all functional departments and lead to organisation-wide concerted efforts for higher supply-chain efficiency.

Recently, the diminished role of marketing has caught the attention of an increasing number of marketing researchers (Sheth & Sisodia, 2006). They showed concerns that the marketing department is no longer solely responsible for the customer-connecting role, since 'marketing is everywhere' within organisations. Formally or informally, marketing is the job of every employee from various functional departments throughout organisations. Verhoef and Leeflang (2009) even proposed the question of whether organisations need a strong marketing department due to the fact that a lot of traditional marketing decisions, such as pricing or distribution, may now be covered by other departments, such as sales and finance. Addressing those concerns, McDonal (2009) emphasises marketing as a process of defining, delivering, and monitoring customer values. Our study shows that the successful execution of such a process depends on the irreplaceable boundary-spanning efforts of marketing managers. Although 'marketing is everywhere', marketing efforts at the individual functional department is only fragmented, and such fragmented efforts provide no value for guiding the strategic decisions of organisations related to defining, delivering, and monitoring customer values. Customer value fulfilment requires boundary-spanning marketing managers to build up both internal and inter-organisational connections of fragmented marketing efforts and strengthen dependencies among various functional departments for cohesive product, service, and financial offerings (Payne & Frow, 2006).

Our study also opens up promising opportunities for future research. Our study focuses on boundary-spanning tasks from the perspective of marketing managers. Recently, a number of researchers have argued that the marketing concept is dispersed among all functional departments within the organisation (e.g. McDonald, 2009). One possible avenue for future research is to examine how managerial roles of other functional departments interact with boundary-spanning roles of marketing managers, and how such interaction impacts market performance of the organisation. Furthermore, the study examines perceived organisational support as a direct predictor of marketing managers' boundary-spanning behaviours. One of the relevant questions is how different forms of organisational control impact marketing managers' execution of those boundary-spanning activities. Researchers have identified various organisational control mechanisms, such as bureaucratic or charismatic control (Desmond, 2004). Is there one specific form of organisational control mechanism that is most suitable for boundary-spanning roles of marketing managers, or is the optimal organisational control mechanism contingent on other factors, such as the line of the industry, market turbulence, or competitive intensity? Empirically examining the impact of organisational structures on marketing managers' boundary-spanning roles will undoubtedly provide more insight on this research issue.

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